

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application Practical

Practical – V

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AIM:Create the following application

A. Create a web application to demonstrate the use of different types of Cookies.

Algorithm:

STEP 1 :- START

STEP 2 :- Initialize the ViewState counter and display/increment it when the View State button is clicked.

STEP 3 :- Store the entered name in a cookie, redirect to the second page, and display the cookie value.

STEP 4 :- STOP

CODE:

webfrom1.aspx.cs:-

```
protected void Page_Load(object sender, EventArgs e)
{
    if (IsPostBack)
    {
        int viewstateval1 = Convert.ToInt32(ViewState["count"]) + 1;
        Label2.Text = "View State =" + viewstateval1.ToString();
    }
    else {
        ViewState["count"] = 1;
    }
}

protected void Button1_Click(object sender, EventArgs e)
{
    Label1.Text = ViewState["count"].ToString();
}

protected void Button2_Click(object sender, EventArgs e)
```

```

{
    HttpCookie h = new HttpCookie("name");
    h.Value = TextBox1.Text;
    Response.Cookies.Add(h);
    Response.Redirect("Webform2.aspx");
}
}

```

webform2.aspx.cs:-

```

protected void Page_Load(object sender, EventArgs e)
{
    if (Request.Cookies["name"] != null) {
        Response.Write("WElcome" + Request.Cookies["name"].Value);
    }
}

```

OUTPUT:

1

View State =2

View state

Ritesh Chaubey

Cookies

WElcomeRitesh Chaubey
Label

B. Create a Product Table in SQL Server and perform Insert Operation using Entity Framework Core.

Algorithm:

STEP 1 :- START

STEP 2 :- Establish a connection to the SQL Server database using Entity Framework Core.

STEP 3 :- Create a Product object and insert it into the Products table.

STEP 4 :- Save the changes and display the message "Product Inserted Successfully".

STEP 5 :- STOP

CODE:

HomeController.cs:

```
using Microsoft.AspNetCore.Mvc;

public class HomeController : Controller
{
    private readonly ApplicationDbContext _context;

    public HomeController(ApplicationDbContext context)
    {
        _context = context;
    }

    public IActionResult Index()
    {
        Product p = new Product();

        p.Name = "Laptop";
        p.Name = "MacBook";

        p.Price = 70000;
        p.Price = 150000;

        _context.Products.Add(p);
        _context.SaveChanges();

        return Content("Product Inserted Successfully.");
    }
}
```

Program.cs:

```
using Microsoft.EntityFrameworkCore;
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.
builder.Services.AddControllersWithViews();

builder.Services.AddDbContext<ApplicationDbContext>(options =>
options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnec
tion")));
```

```

var app = builder.Build();

app.UseStaticFiles();
app.UseRouting();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}")
    .WithStaticAssets();

```

```
app.Run();
```

appsetting.json:

```

{
  "ConnectionStrings": {
    "DefaultConnection": "Server = (localdb)\\MSSQLLocalDB;Database=ProductDB ,
Trusted_Connection = True; TrustServerCertificate = True ;"
  }
}

```

AppDbContext.cs:

```

using Microsoft.EntityFrameworkCore;

public class AppDbContext : DbContext
{
    public AppDbContext(DbContextOptions<AppDbContext> options) :
    base(options)
    {

    }

    public DbSet<Product> Products { get; set; }
}

```

Product.cs:

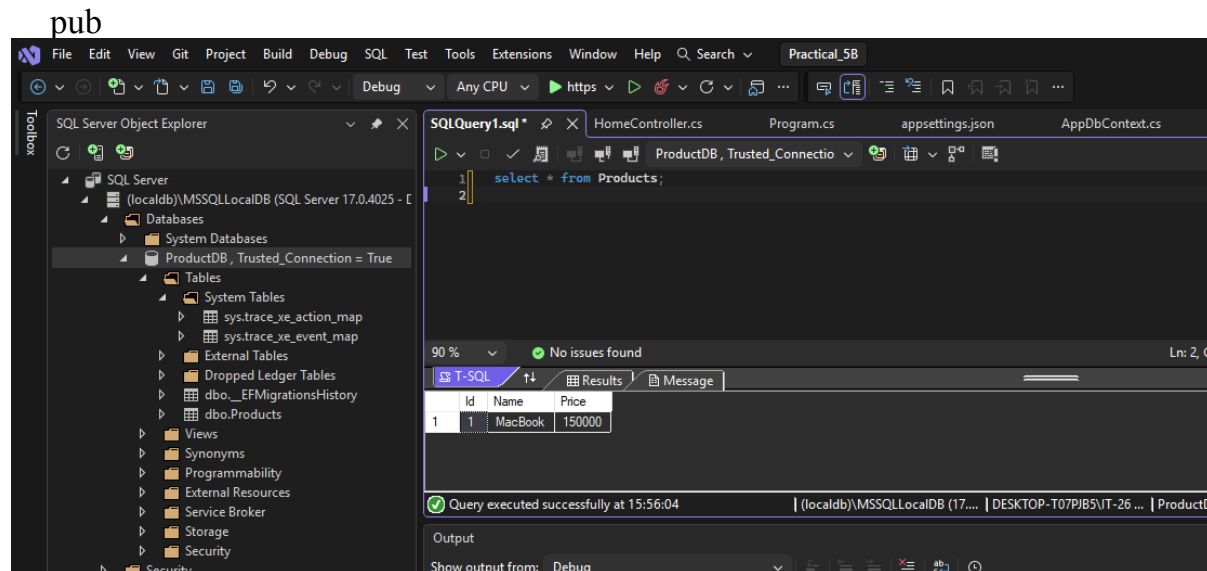
```

public class Product
{
    public int Id { get; set; }
    public string Name { get; set; }
    int Price { get; set; }
}

```

OUTPUT:

```
Product Inserted Successfully.
```



C.Develop a web application to perform CRUD Operations (Create, Read, Update, Delete) on the Employee table using EF Core.

Algorithm:

STEP 1 :- START

STEP 2 :- Connect to the SQL Server database.

STEP 3 :- Perform CRUD operations (Create, Read, Update, Delete) on Employee records.

STEP 4 :- Display the updated Employee information to the user.

STEP 5 :- STOP

CODE:

AppDbContext.cs:-

using Microsoft.EntityFrameworkCore;

namespace WebApplication11.Data

```
{
    public class AppDbContext:DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options) { }
        public DbSet<Models.Employee> Employees { get; set; }
    }
}
```

Employee.cs:-

namespace WebApplication11.Models

```
{
    public class Employee
    {
        public int Id { get; set; }
        public String Name { get; set; }
        public String Departement { get; set; }
        public double salary { get; set; }
    }
}
```

Appsetting.json:-

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "MyConnection":
"Server=(localdb)\\MSSQLLocalDB;Database=EmployeeDB;Trusted_Connection=True;TrustServerCertificate=True",
    "AppDbContext":
"Server=(localdb)\\mssqllocaldb;Database=AppDbContext;Trusted_Connection=True;MultipleActiveResultSets=true"
  }
}
```

Program.cs:-

```
using Microsoft.EntityFrameworkCore;
using WebApplication11.Data;
```

```
builder.Services.AddControllersWithViews();
builder.Services.AddDbContext<AppDbContext>(options =>
options.UseSqlServer(builder.Configuration.GetConnectionString("MyConnection")));
```

OUTPUT:

WebApplication11 Home Privacy			
Index			
Create New			
Name	Departement	salary	
Ritesh	It	120000	Edit Details Delete

Delete

Are you sure you want to delete this?

Practical_5C.Models.Employee

Name	Reze
Dept	Anime
salary	5000

Delete | [Back to List](#)

D.Authentication and Authorization in ASP.NET Core.

Algorithm:

STEP 1 :- START

STEP 2 :- Accept the username and password from the user.

STEP 3 :- Verify the entered credentials and authenticate the user using cookie authentication.

STEP 4 :- Grant access to the secure page if login is successful; otherwise, display an error message.

CODE:

Program.cs:-

```
using Microsoft.AspNetCore.Authentication.Cookies;
```

```
var builder = WebApplication.CreateBuilder(args);
```

```
builder.Services.AddControllersWithViews();
```

```
builder.Services.AddAuthentication(CookieAuthenticationDefaults.AuthenticationScheme)
.AddCookie(options =>
```

```

{
    options.LoginPath = "/Home/Index";
});

builder.Services.AddAuthorization();

var app = builder.Build();

app.UseStaticFiles();

app.UseRouting();

app.UseAuthentication();
app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");

app.Run();

```

HomeController.cs:-

```

using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.Authorization;
using Microsoft.AspNetCore.Authentication;
using Microsoft.AspNetCore.Authentication.Cookies;
using System.Security.Claims;

namespace AuthDemo.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index()
        {
            return View();
        }

        [HttpPost]
        public async Task<IActionResult> Index(string username, string password)
        {
            if (username == "admin" && password == "123")
            {

```



```

        var claims = new List<Claim>
        {
            new Claim(ClaimTypes.Name, username)
        };

        var identity = new ClaimsIdentity(claims,
            CookieAuthenticationDefaults.AuthenticationScheme);

        await HttpContext.SignInAsync(
            CookieAuthenticationDefaults.AuthenticationScheme,
            new ClaimsPrincipal(identity));

        return RedirectToAction("Secure");
    }

    ViewBag.Message = "Invalid Username or Password";
    return View();
}

[Authorize]
public IActionResult Secure()
{
    return Content("Welcome Ritesh Chaubey! Login Successful.");
}

public async Task<IActionResult> Logout()
{
    await HttpContext.SignOutAsync(
        CookieAuthenticationDefaults.AuthenticationScheme);

    return RedirectToAction("Index");
}
}
}

```

Index.cshtml:-

```
<h2>Login</h2>
```

```
<form method="post">
```

Username :

```
<input type="text" name="username" /><br /><br />
```

Password :

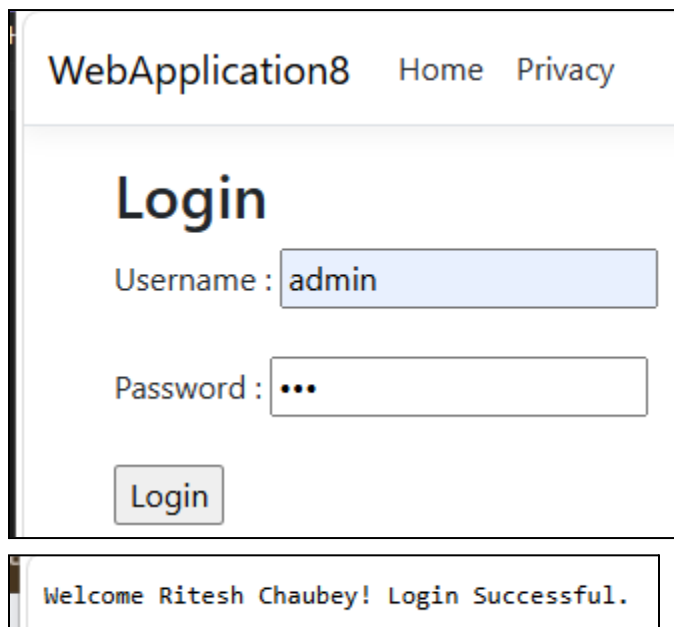
```
<input type="password" name="password" /><br /><br />
```

```
<input type="submit" value="Login" />
```

```
</form>
```

```
<p style="color:red">  
    @ViewBag.Message  
</p>
```

OUTPUT:



WebApplication8 Home Privacy

Login

Username : admin

Password : ...

Login

Welcome Ritesh Chaubey! Login Successful.